

How to implement UP or Down Counter by Combinations of Clock & Outputs

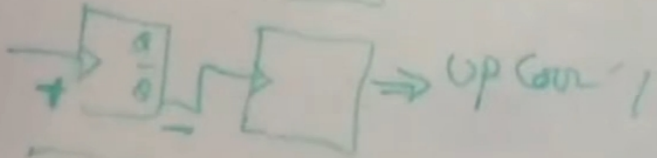
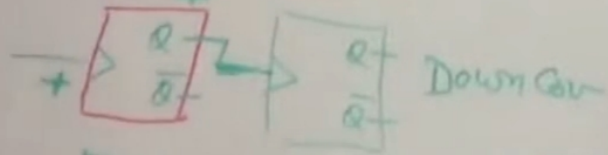
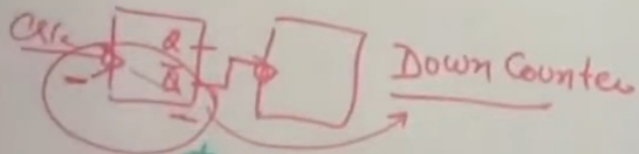
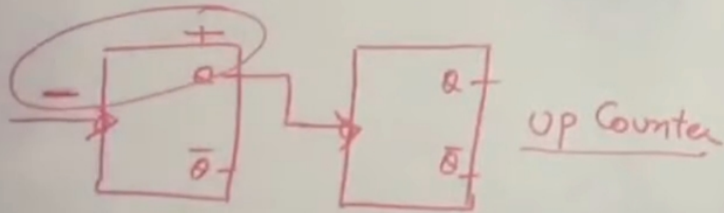
$\boxed{- \& +ve}$ $\rightarrow$ U_P

3-Bit UP-DOWN RIPPLE COUNTER

- ① -ve Edge Trig FF $\rightarrow Q$ is clock $\Rightarrow$ UP Counter
- ② -ve Edge Trig FF $\rightarrow \bar{Q}$ is clock $\Rightarrow$ Down Counter
- ③ +ve Edge Trig FF $\rightarrow Q$ is clock $\Rightarrow$ Down Counter
- ④ +ve Edge Trig FF $\rightarrow \bar{Q}$ is clock $\Rightarrow$ UP Counter

$\Rightarrow$ Both Up & Down modes are combined

$\Rightarrow$ A mode Control I/P (M) is used to select

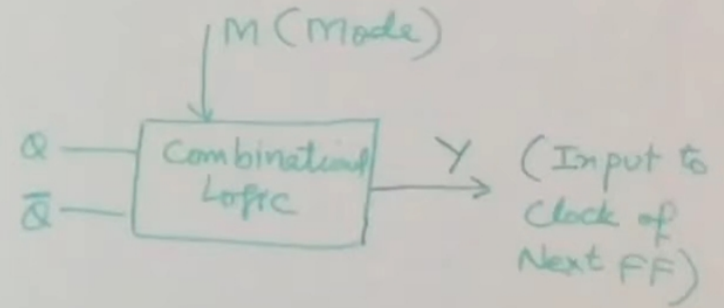
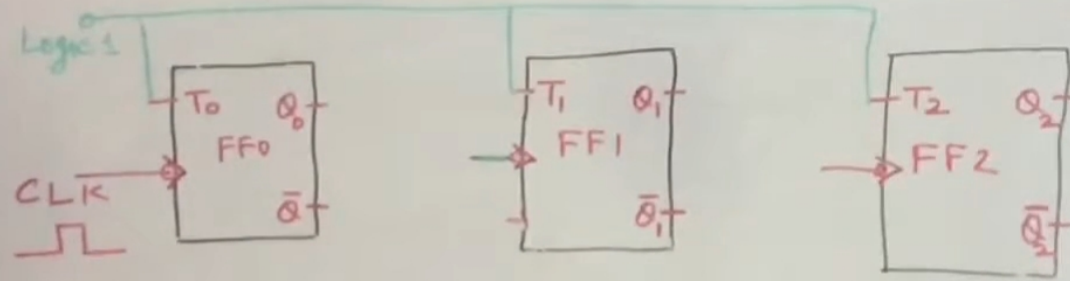


Block diagram in which we will insert Logic of I/P M :-

3-Bit UP-DOWN RIPPLE COUNTER

- ⇒ Both Up & Down modes are combined
- ⇒ A mode Control I/P (M) is used to select

M = 0 — Up Counting
M = 1 — Down Counting



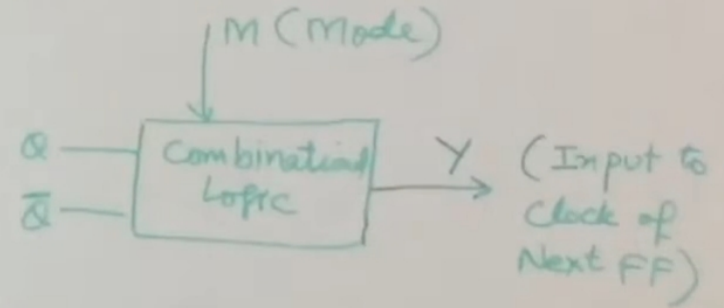
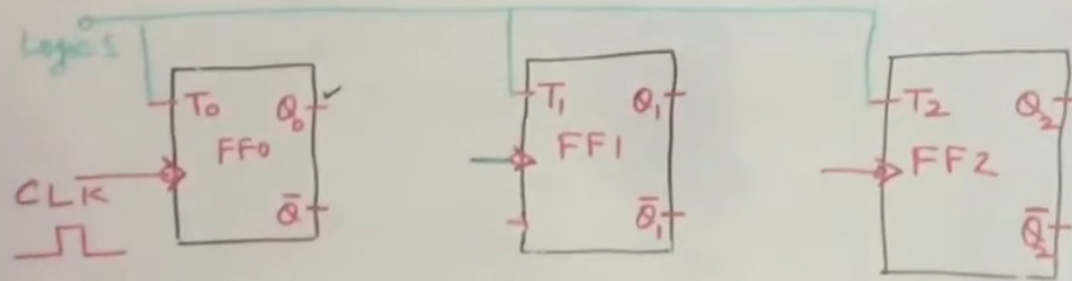
State Diagram for M along with Q & $\bar{Q}$ inputs & Output Y :-

3-Bit UP-DOWN RIPPLE COUNTER

⇒ Both Up & Down modes are combined

⇒ A mode Control I/P (M) is used to select

M = 0 — Up Counting
M = 1 — Down Counting



	M	Q	$\bar{Q}$	Y
Up Counting	0	0	1	0
	0	0	1	0
	0	1	0	1
	0	1	0	1
Down Counting	1	0	0	0
	1	0	1	1
	1	1	0	0
	1	1	1	1

K-Map for input M along with Q & Q̄ :-

3-Bit UP-DOWN RIPPLE COUNTER

Handwritten K-Map for a 3-bit up-down ripple counter. The map is a 2x4 grid with inputs M and Q as variables.

M \ Q	00	01	11	10
0	0	1	0	0
1	0	1	1	1

Red circles highlight the prime implicants: a vertical circle around the 1s in the Q=01 column, and a horizontal circle around the 1s in the M=1 row.

$$Y = \underbrace{\bar{M} \cdot Q}_{\text{AND}} + \underbrace{M \cdot \bar{Q}}_{\text{AND}}$$

Final Ckt. for 3-bit Up & Down Async. counter :-

